

Open Science & FAIR

Introduction to Data Management Practices course

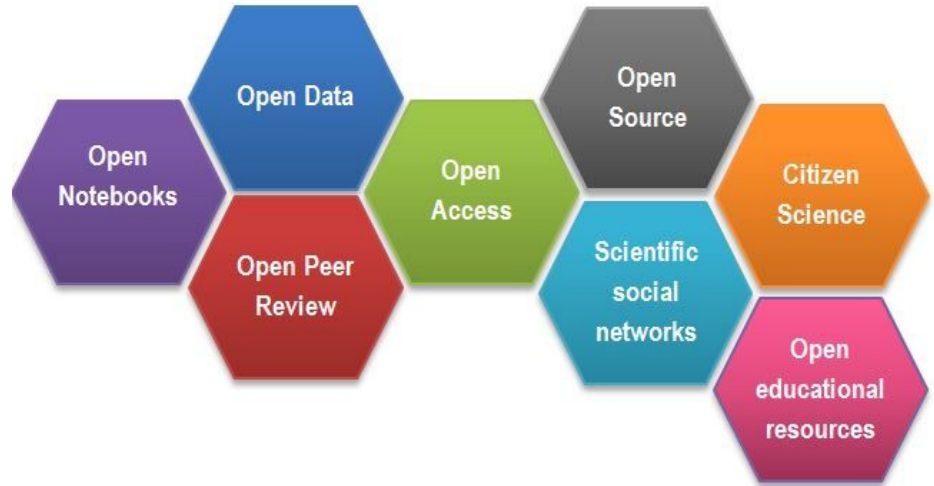
NBIS DM Team

data-management@scilifelab.se

Open Science

Make scientific research and its dissemination accessible to all levels of society.

- Open methodology
- **Open source**
- *Open data*
- Open access
- Open peer review
- Open educational resources



"Open Science facets as a beehive" by Gema
Bueno de la Fuente licenced under [CC-BY](#)

UNESCO Open Science Recommendation & Toolkit

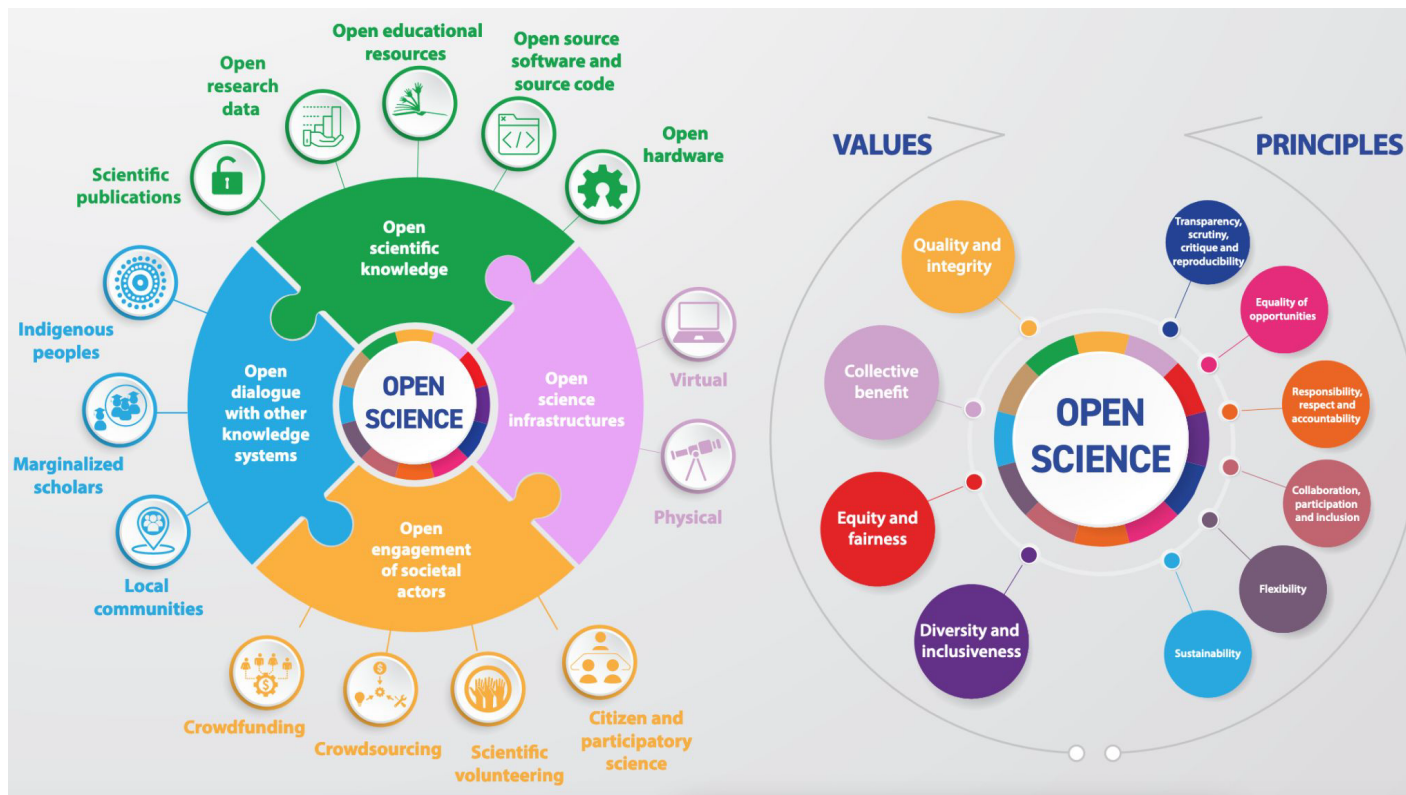


Illustration from [UNESCO open science toolkit: contents](#) under a [Creative Commons Attribution Share-Alike 3.0 License](#)

**What do you think are
reasons for Open Data?**

Open Data

- Democracy and transparency
 - Publicly funded research data should be accessible to all
 - Published results and conclusions should be possible to check by others
- Research
 - Enables others to combine data, address new questions, and develop new analytical methods
 - Reduce duplication and waste
- Innovation and utilization outside research
 - Public authorities, companies, and private persons outside research can make use of the data
- Citation
 - Citation of data will be a merit for the researcher that produced it



Ethical?

Doing “sloppy” science & not being open and transparent

Waste of resources

Contributing to the current research credibility crisis

Contributing to the current reproducibility crisis

Harming the profession

Harming public trust in research

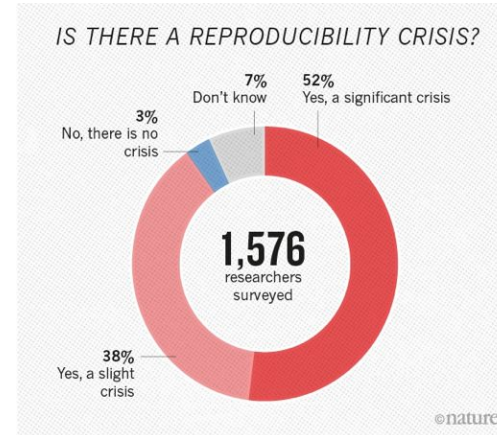
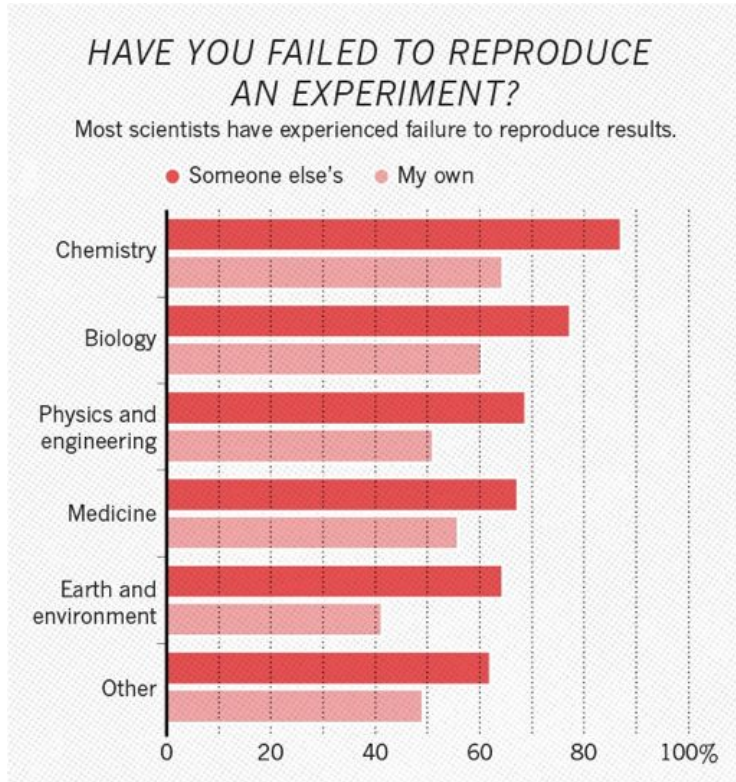


Crisis?

Do you think we have a **credibility** and/or **reproducibility** crisis?

If so, what are some of its causes?

A reproducibility crisis



[1] "1,500 scientists lift the lid on reproducibility". Nature. 533: 452–454

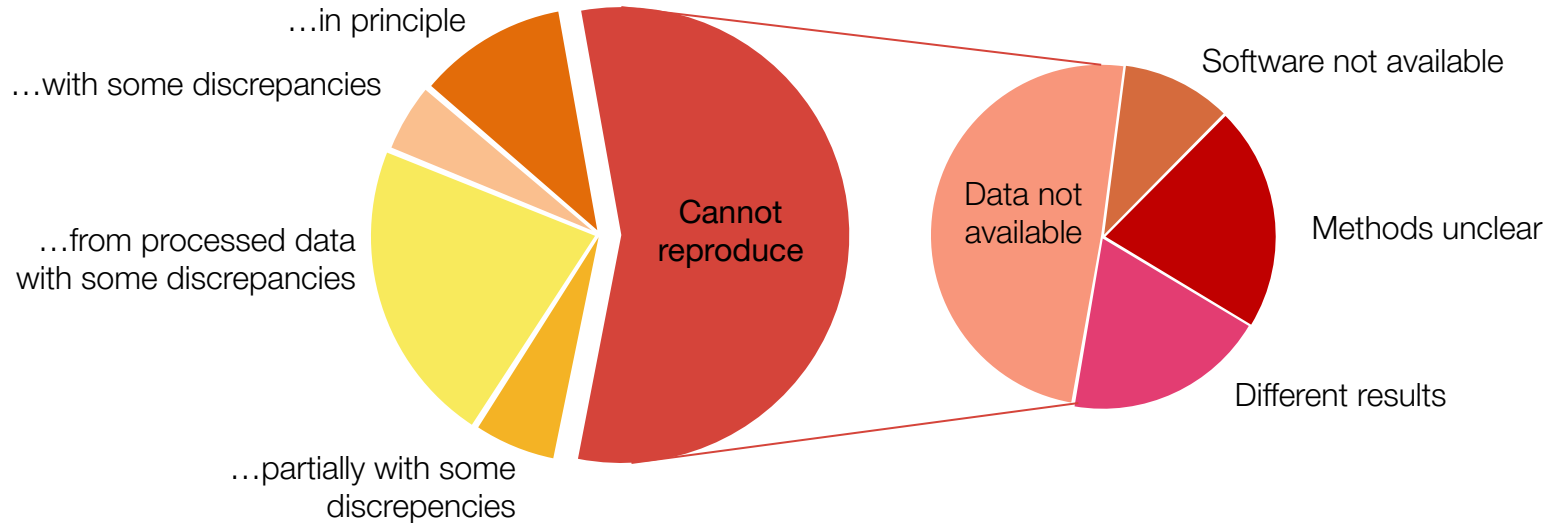
[2] Begley, C. G.; Ellis, L. M. (2012). "Drug development: Raise standards for preclinical cancer research". Nature. 483 (7391): 531–533.



A reproducibility crisis

Reproduction of data analyses in 18 articles on microarray-based gene expression profiling published in Nature Genetics in 2005–2006:

Can reproduce...



Summary of the efforts to replicate the published analyses.

Adopted from: Ioannidis et al. Repeatability of published microarray gene expression analyses. *Nature Genetics* 41 (2009) doi:10.1038/ng.295

Data Management Snafu



<https://www.youtube.com/watch?v=N2zK3sAtr-4>

FAIR

- To be useful for others data should be:
 - **FAIR** - Findable, Accessible, Interoperable, and Reusable
... for both Machines and Humans

Wilkinson, Mark et al. “The FAIR Guiding Principles for scientific data management and stewardship”. Scientific Data 3, Article number: 160018 (2016) <http://dx.doi.org/10.1038/sdata.2016.18>



DOI: 10.1038/sdata.2016.18

Box 2 | The FAIR Guiding Principles

To be Findable:

- F1. (meta)data are assigned a globally unique and persistent identifier
- F2. data are described with rich metadata (defined by R1 below)
- F3. metadata clearly and explicitly include the identifier of the data it describes
- F4. (meta)data are registered or indexed in a searchable resource

To be Accessible:

- A1. (meta)data are retrievable by their identifier using a standardized communications protocol
- A1.1 the protocol is open, free, and universally implementable
- A1.2 the protocol allows for an authentication and authorization procedure, where necessary
- A2. metadata are accessible, even when the data are no longer available

To be Interoperable:

- I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. (meta)data use vocabularies that follow FAIR principles
- I3. (meta)data include qualified references to other (meta)data

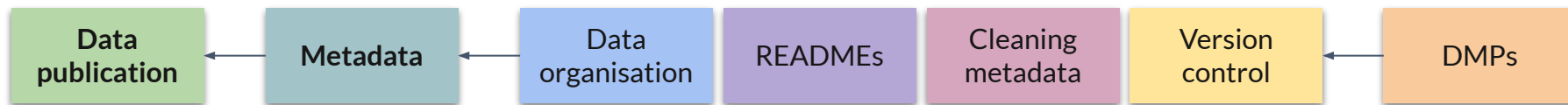
To be Reusable:

- R1. (meta)data are richly described with a plurality of accurate and relevant attributes
- R1.1. (meta)data are released with a clear and accessible data usage license
- R1.2. (meta)data are associated with detailed provenance
- R1.3. (meta)data meet domain-relevant community standards



Findable

- Data have a **globally unique persistent identifier**
 - *e.g. a DOI, database accession number, etc*
- Data are described by **metadata**
 - *Information that explains the data*
- Data and metadata are findable in a **search resource**
 - *There must be ways of searching for the data*



Accessible

- Data is retrievable through a **standardised communication protocol** (open, free, allowing authentication & authorisation where necessary)
 - *e.g. http, sftp, etc*
- Metadata are accessible, **even if data is no longer available**
 - *Information about the data can be found even if data is no longer available*



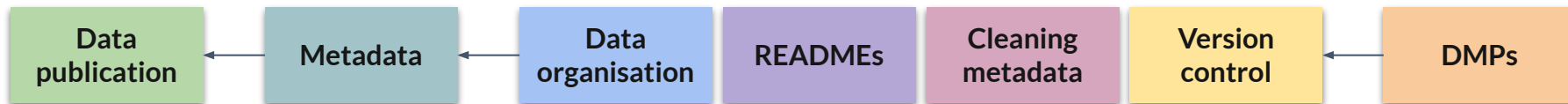
Interoperable

- Metadata use a formal, accessible, shared **language for knowledge representation**
 - *Metadata is available in a form that even a computer can make use of*
- Metadata use **vocabularies** that follow the FAIR principles
 - *Standardised ways of capturing information about the data (that are in themselves FAIR)*
- Metadata include qualified **references** to other metadata
 - *If the data relies on other data, there must be links to those*



Reusable

- Data have a clear **data usage license**
 - *It is obvious under what conditions the data can be reused*
- Metadata are associated with **detailed provenance**
 - *The metadata is detailed enough to understand for what research questions it is relevant to reuse*
- Metadata meet domain-relevant community **standards**
 - *Metadata is described according to existing standards in the research field*



FAIR

- Both humans and machines are intended users of data
- The principles are not necessarily about *open* data
 - “As open as possible, as closed as necessary”
- FAIRness is not something absolute
 - Different levels of FAIR maturity
- FAIR does not enforce any particular technical standards

Metadata?

The data about the data (or anything really)

Example: Spleen samples from mice

organism	strain	age	time unit	developmental stage	sex	organism part	individual	genotype
Mus musculus	C57BL/6J x 129Sv	12	week	adult	male	spleen	287 - 2	HellsF/F Mb1Cre+
Mus musculus	C57BL/6J x 129Sv	12	week	adult	male	spleen	287 - 2	HellsF/F Mb1Cre+
Mus musculus	C57BL/6J x 129Sv	15	week	adult	female	spleen	278 - 6	HellsF/F Mb1Cre+
Mus musculus	C57BL/6J x 129Sv	15	week	adult	female	spleen	278 - 6	HellsF/F Mb1Cre+

Simple FAIR data example


ArrayExpress

Examples: [E-MEXP-31](#), [cancer](#), [p53](#), [Geuvadis](#)

[Home](#) | [Browse](#) | [Submit](#) | [Help](#) | [About ArrayExpress](#)

[Contact Us](#) | [Login](#)

Search results for **"organism:mus_musculus" AND exptype:"RNA-seq of coding RNA" AND expdesign:"time series" AND "organism part:liver"**

6 experiments

Accession	Title	Type	Organism	Assays	Released	Processed	Raw	Atlas
E-MTAB-10239	scRNA-seq of murine mucosal associated invariant T (MAIT) cells after Francisella tularensis infection	RNA-seq of coding RNA from single cells	Mus musculus	3	19/05/2021	-		-
E-MTAB-7054	Transcriptional profiling of hepatic stellate cells (HSCs) isolated from Western diet/high fructose-fed C57BL6/J mice, carbon tetrachloride (CCl4)-treated C57BL6/J mice, and of murine HSCs differentiated in vitro	RNA-seq of coding RNA	Mus musculus	53	07/04/2019			-
E-MTAB-6435	Transcriptome profiling of liver samples of C/EBPβ ΔuORF mice	RNA-seq of coding RNA	Mus musculus	24	16/01/2019	-		
E-MTAB-7020	RNA-seq study on time of day specific Glucocorticoid action in mouse liver and lung tissues	RNA-seq of coding RNA	Mus musculus	32	13/11/2018	-		-
E-MTAB-7017	RNASeq data analysis of wild type and reverb alpha knockout cells from mouse liver, at different time points, with or without DEX treatment	RNA-seq of coding RNA	Mus musculus	40	13/11/2018	-		
E-MTAB-2351	RNA-seq of Sod1 deficient and wild type mice after lymphocytic choriomeningitis virus (LCMV) infection	RNA-seq of coding RNA	Mus musculus	18	17/11/2015	-		-

Picture source: [ArrayExpress @ EMBL-EBI](#)



FAIR at source?



Retroactively?

Good Data Management Practices

- **Data Management Plans**, to do your thinking ahead of time
- **Using standard metadata descriptions**, to clearly define your data
- **Organising your outputs**, so you and others can understand what you have done
- **Use versioning control** to keep track of changes you do
- **Clean up metadata and data** to be consistent with the standards you have chosen
- **Submit your data to international public repositories**, so others can find and reuse your data

What data management practices do you apply in your research projects today?

	Ad Hoc	One-Time	Active and Informative	Optimized for Re-Use
Planning your project	When it comes to my data, I have a "way of doing things" but no standard or documented plans.	I create some formal plans about how I will manage my data, but I generally don't refer back to them.	I develop detailed plans about how I will manage my data that I actively revisit and revise over the course of a project.	I design my plans for managing data to streamline future use by myself or others.
Organizing your data	I don't follow a consistent approach for keeping my data organized, so it often takes time to find things.	I have an approach for organizing my data, but I only put it into action after my project is complete.	I have an approach for organizing my data that I implement prospectively, but it not necessarily standardized.	I organize my data to the so that others can navigate, understand, and use it without me being present.
Saving and backing up your data	I decide what data is important while I am working on it and typically save it in a single location.	I know what data needs to be saved and I back it up after I'm done working on it to reduce the risk of loss.	I have a system for regularly saving important data while I am working on it. I have multiple backups.	I save my data in a manner and location designed maximize opportunities for re-use by myself and others.
Getting your data ready for analysis	I don't have a standardized or well documented process for preparing my data for analysis.	I have thought about how I will need to prepare my data, but I handle each case in a different manner.	My process for preparing data is standardized and well documented.	I prepare my data in such a way as to facilitate use by both myself and others in the future.
Analyzing your data and handling the outputs	I often have to redo my analyses or examine their products to determine what procedures or parameters were applied.	After I finish my analysis, I document the specific parameters, procedures, and protocols applied.	I regularly report the specifics of both my analysis workflow and decision making process while I am analyzing my data.	I have ensured that the specifics of my analysis workflow and decision making process can be put into action by others.
Sharing and publishing your data	I share the results of my research, but generally I do not share the underlying data.	I share my my data only when I'm required to do so or in response to direct requests from other researchers.	I regularly share the data that underlies my results and conclusions in a form that enables use by others.	Because of my excellent data management practices, I am able to efficiently share my data whenever I need to with whomever I need to.

Borghi, J. et al (2018). *Support your Data.*
<https://doi.org/10.3897/rio.4.e26439>

Voting

	Ad Hoc	One-Time	Active and Informative	Optimized for Re-Use
Planning your project	When it comes to my data, I have a "way of doing things" but no standard or documented plans.	I create some formal plans about how I will manage my data, but I generally don't refer back to them.	I develop detailed plans about how I will manage my data that I actively revisit and revise over the course of a project.	I design my plans for managing data to streamline future use by myself or others.
Organizing your data	I don't follow a consistent approach for keeping my data organized, so it often takes time to find things.	I have an approach for organizing my data, but I only put it into action after my project is complete.	I have an approach for organizing my data that I implement prospectively, but it not necessarily standardized.	I organize my data to the so that others can navigate, understand, and use it without me being present.
Saving and backing up your data	I decide what data is important while I am working on it and typically save it in a single location.	I know what data needs to be saved and I back it up after I'm done working on it to reduce the risk of loss.	I have a system for regularly saving important data while I am working on it. I have multiple backups.	I save my data in a manner and location designed maximize opportunities for re-use by myself and others.
Getting your data ready for analysis	I don't have a standardized or well documented process for preparing my data for analysis.	I have thought about how I will need to prepare my data, but I handle each case in a different manner.	My process for preparing data is standardized and well documented.	I prepare my data in such a way as to facilitate use by both myself and others in the future.
Analyzing your data and handling the outputs	I often have to redo my analyses or examine their products to determine what procedures or parameters were applied.	After I finish my analysis, I document the specific parameters, procedures, and protocols applied.	I regularly report the specifics of both my analysis workflow and decision making process while I am analyzing my data.	I have ensured that the specifics of my analysis workflow and decision making process can be put into action by others.
Sharing and publishing your data	I share the results of my research, but generally I do not share the underlying data.	I share my my data only when I'm required to do so or in response to direct requests from other researchers.	I regularly share the data that underlies my results and conclusions in a form that enables use by others.	Because of my excellent data management practices, I am able to efficiently share my data whenever I need to with whomever I need to.

https://bit.ly/support_your_data_rubric

The Political landscape



The Political Landscape

- Policymakers are **pushing for research data to be made available** as openly as possible
- Big investments are being made in **infrastructure and skills for data sharing and reuse**



Policy landscape

- [UNESCO's Recommendation on Open Science](#)
- EU
 - Open Science Policy, Directives
 - European Research Council, Horizon Europe
 - EOSC
- Swedish research policy bills 2016, 2020, 2024
 - Transition to open research data by 2026 & push for FAIR
 - Government assignments to KB & VR
- [SUHF national roadmap for open science](#)
- [University policies](#)
- [National guidelines for open science](#) (KB)

Open Science

FAIR

*“As open as possible,
as closed as necessary”*

*“FAIR [...] open data sharing should
become the default [...]”*

*“[...] research results [...] meet the
FAIR principles and can be reused
by everyone”*



National Guidelines for Open Science



National guidelines for promoting open science in Sweden

15 januari 2024

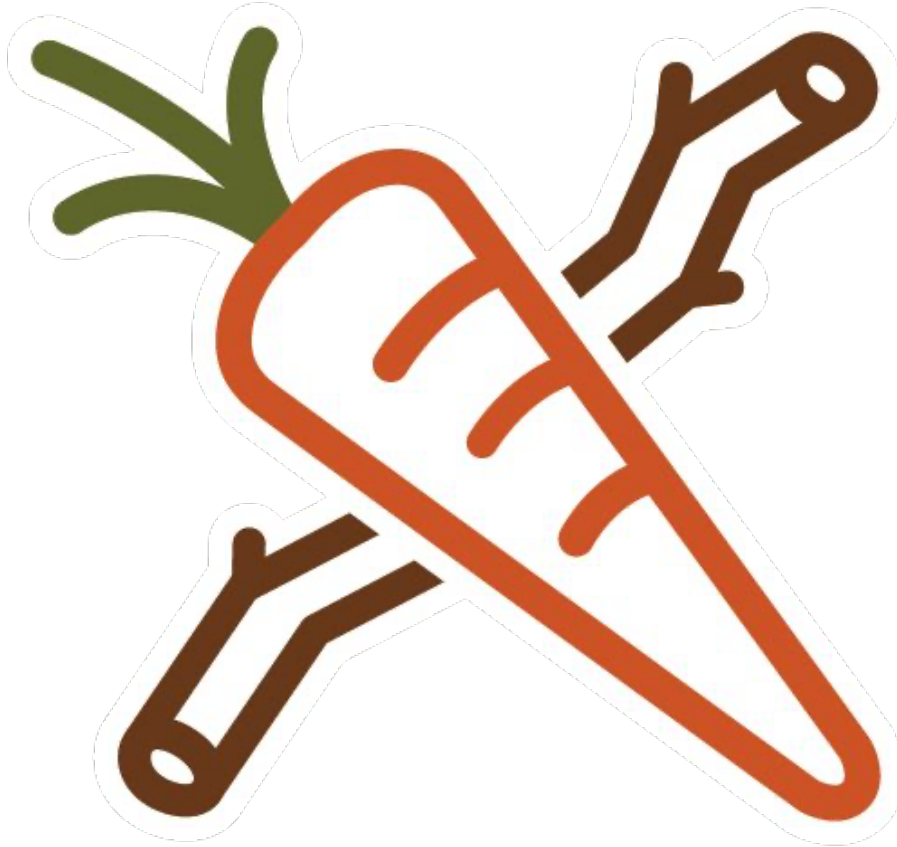
 Open Science (In English)

On behalf of the Swedish government, the National Library of Sweden (Kungliga biblioteket, KB) has developed national guidelines for open science. The guidelines are intended to provide support and guidance to actors in Sweden who have an important role to play in the transition to open science.

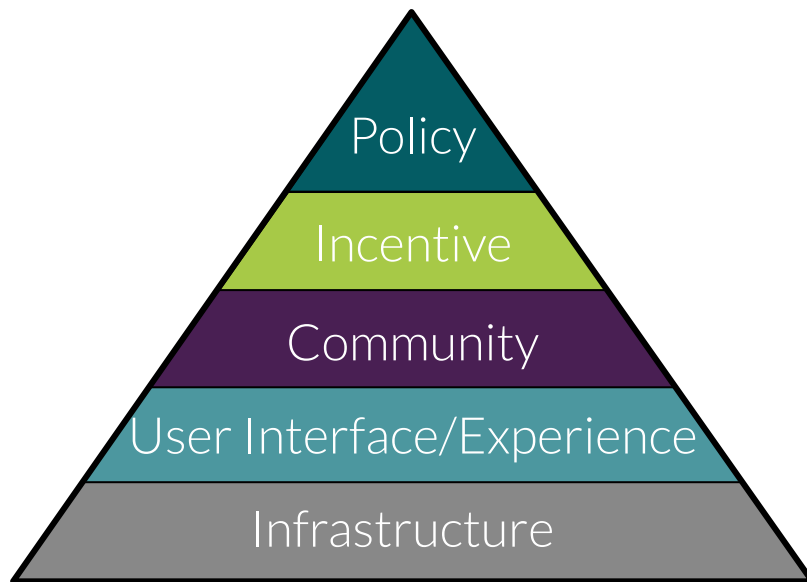


- **Open Access to Scholarly Publications:** Research publications freely available without any subscription or payment barriers
- **Open Access to Research Data:** Availability/accessibility of research data for reproducibility, to further scientific inquiry
- **Open Research Methods:** Transparency of research methodologies to facilitate replication and validation of research findings
- **Open Educational Resources:** Freely accessible educational materials to enhance learning and teaching practices
- **Public Engagement in Science:** Public, community, citizen science to increase public understanding and trust in science
- **Infrastructures Supporting Open Science:** Develop/support infrastructures for open science ecosystem, including repositories and data management systems

Motivators



Open Science: from policy to practice



Make it required

Make it rewarding

Make it normative

Make it easy

Make it possible

Good Data Management Practices

- **Data Management Plans**, to do your thinking ahead of time
- **Using standard metadata descriptions**, to clearly define your data
- **Organising your outputs**, so you and others can understand what you have done
- **Use versioning control** to keep track of changes you do
- **Clean up metadata and data** to be consistent with the standards you have chosen
- **Submit your data to international public repositories**, so others can find and reuse your data